Note

That the control variable is **i**, which is initialized to 10, the condition is tested **i<=15,** if true value of **i** gets printed, then the control variable i gets updated as **i=i+1** (this can also be written as i +=1 using shorthand assignment operator). When **i** becomes **16**, the condition is tested False and this will terminate the loop.

Note

print can have end, sep as parameters. *end* parameter can be used when we need to give any escape sequences like '\t' for tab, '\n' for new line and so on. *sep* as parameter can be used to specify any special characters like, (comma) ; (semicolon) as separator between values (Recall the concept which you have learnt in previous chapter about the formatting options in print()).

Following is an example for using else part within while loop.

Example 6.7: program to illustrate the use of while loop - with else part

```
i=10                                          # intializing part of the control variable
while (i<=15):                                # test condition
      print (i,end='\t')                      # statements - block 1
      i=i+1                                   # Updation of the control variable
else:
    print ("\nValue of i when the loop exit ",i)
```

Output: 1

```
10   11    12    13    14    15
Value of i when the loop exit  16
```

(ii) for loop

for loop is the most comfortable loop. It is also an entry check loop. The condition is checked in the beginning and the body of the loop(statements-block 1) is executed if it is only True otherwise the loop is not executed.

Syntax:

```
for counter_variable in sequence:
      statements-block 1
[else:                          # optional block
      statements-block 2]
```

The *counter_variable* mentioned in the syntax is similar to the control variable that we